Title: Care Transition Efficiency & Placement Outcome Analytics

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1. **Problem Statement**

   The UAC (Unaccompanied Alien Children) Program tracks children as they move through a care pipeline: from CBP custody, to HHS care, and finally to a vetted sponsor. Currently, the system monitors capacity (how many children are in custody), but it lacks tools to measure process efficiency. Without knowing how fast transfers and discharges happen, system bottlenecks remain hidden.

2. **Proposed Solution & Methodology**

   To address this, I built an interactive Streamlit web application using Python. This dashboard reframes the data to focus on pipeline flow and efficiency. Using synthetic data to model real-world conditions, the dashboard features:

- A flow visualization diagram mapping the volume of children across stages.
- Time-series charts tracking transfer and discharge efficiency.
- A backlog detection system that flags periods where intakes outpace successful discharges.

3. Key Metrics Monitored

- **Transfer Efficiency Ratio:** Measures the speed of moving children from CBP to HHS.
- **Discharge Effectiveness:** Measures the success rate of placing children from HHS to sponsors.
- **Pipeline Throughput:** The overall system movement from start to finish.
- **Backlog Accumulation Rate:** Calculates the severity of delays and piled-up cases.

4. Conclusion

By analyzing how effectively children move through the care pipeline rather than just counting them, this dashboard provides actionable insights. It allows government stakeholders to spot delays early, improve reunification timelines, and ultimately strengthen child welfare outcomes.